

Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

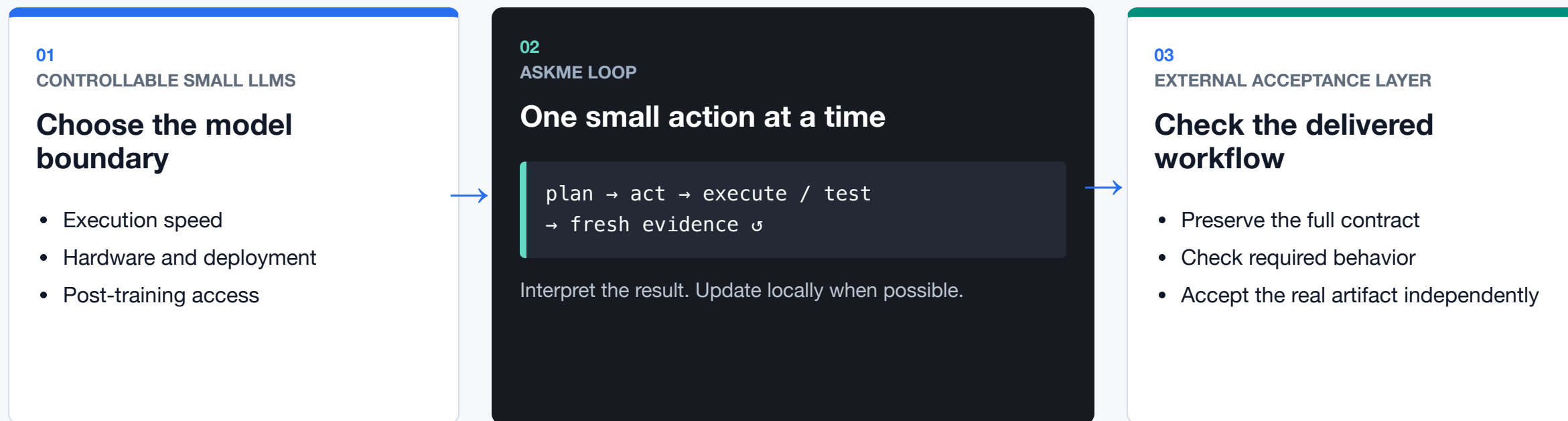
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AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one JSON action → execution evidence ☞



The bridge: fixed action vocabulary + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

A different target was chosen

```
cc -o /tmp/test main.c && /tmp/test
```

The combined command exited 0

The retained record does not independently prove stdout or source contents.

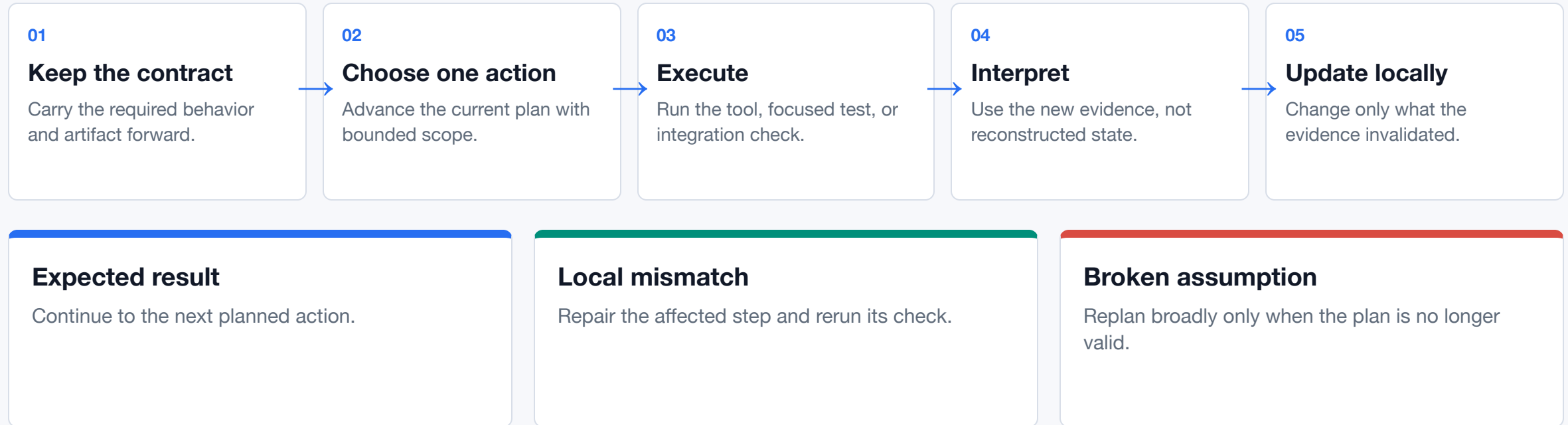
Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion ≠ accepted artifact.

Current limit: acceptance scored this run after completion; the failure was not returned for recovery.

Hypothesis: update only what fresh evidence invalidates



Trajectory goal—not yet isolated: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

All four variants ran; one deliverable was rejected

4 hosted variants × 2 deliberately simple checks × 1 unseeded trajectory per cell

8 / 8 reported complete

7 / 8 artifact accepted

Compatibility smoke · n=1/cell · descriptive only

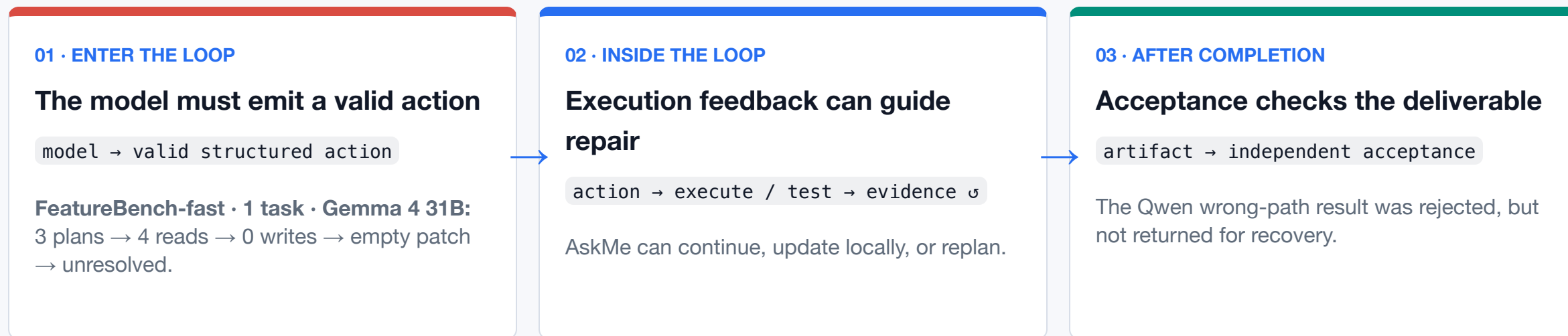
MODEL · SHAPE	ARTIFACT BUILD · OBSERVED TRAJECTORY	SANITY REPAIR · OBSERVED TRAJECTORY
Gemma 4 26B A4B ♦ MoE · 25.2B total / 3.8B active	ACCEPTED 603.6s · 19 steps · 19.5k tok 1 full / 1 local replan	ACCEPTED 20.0s · 7 steps · 4.5k tok 0 full / 0 local replans
Gemma 4 31B • Dense · 30.7B parameters	ACCEPTED 66.5s · 4 steps · 4.4k tok 0 full / 0 local replans	ACCEPTED 22.4s · 5 steps · 4.2k tok 0 full / 1 local replan
Qwen3.6-27B • Dense · 27B parameters	ACCEPTED 47.9s · 6 steps · 5.3k tok 0 full / 0 local replans	ACCEPTED 23.0s · 6 steps · 4.1k tok 0 full / 0 local replans
Qwen3.6-35B-A3B ♦ MoE · 35B total / 3B active	NOT ACCEPTED 17.7s · 3 steps · 3.5k tok wrong artifact path	ACCEPTED 11.8s · 4 steps · 2.5k tok 0 full / 0 local replans

Supported
All four reported completion on both checks; acceptance rejected one wrong deliverable.

Not a clean model comparison
No pair isolates size: dense/MoE shape, active compute, run order, and trajectories changed. No Qwen-vs-Gemma, larger-vs-smaller, speed, reasoning, or reliability inference.

Feedback works only when actions enter and failures return

One limit appeared before execution; another after reported completion.



External boundary probe—not a score: the FeatureBench canary exposed a 512-token structured-action bottleneck before a patch existed. One task cannot separate model capability from this model–harness interface.

Requalified Aug 1 · revision-3 interface · one-task canaries, not scores: sentinel write transport + 4096/8192 budgets + write forcing moved both cells from empty patch to an applied patch — Gemma 4 31B at 11/13 F2P (84.62%), exactly the preregistered pi-ablation ceiling with the identical two failing tests; Qwen3.6-27B at 7/13 (53.85%) vs its 76.92% ceiling. Both still uncanaried by Cerebras, but the still a Green test by Cerebras (Gemma 4 31B at 11/13 F2P) in SiliconFlow for the H/hi results — seeing a stack-overflow-related error in model file 18 times, ran zero tests. (no local claim) · 3 frozen Codex P2 caveats on write-forcing counts · records in tests/featurebench/results/

Promising for bounded loops. Feature readiness is still open.

Observed

Simple smoke: 7 / 8 artifacts accepted. Feature canary: empty patch → applied at 11/13 F2P, the preregistered ceiling, after the revision-3 interface (one task, changed serving stack — not a score).

Supported

The harness exposed two distinct limits: a wrong deliverable and an action that never reached execution. The revision-3 transport moved that action into execution — and exposed a commit-without-validate rewrite loop.

Still open

Feature-scale reliability beyond one canary; stopping the rewrite loop — validate-after-write pressure, rewrite damping, and an unvalidated-write replan flag are in progress, not merged; reasoning impact; Qwen vs Gemma; size, architecture, and local performance.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries — not a leaderboard.

	AskMe	pi	OpenHands
ACTION SURFACE	6 fixed JSON actions; exactly one action per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible <code>Action</code> → <code>Observation</code> tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	<code>done</code> → conditional fail-open validation; held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	<code>finish</code> signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.